Corrected MCQs

1. **B.** The multiplication vector t is a director; the first triplet designates a point on the right.
2. **D.** Take $t = 1$ simultaneously fulfils the three coordinates. Only one correct coordinate does It's not enough.
3. **A.** In an orthonormal coordinate system, the coefficients of x, y, z give a normal vector. constant is not its third coordinated.
4. **C.** These two conditions characterize the project. M is in the plane, the null vector is suitable and $H = M$.
5. **B.** The new point is on the right, and 3 u is a non-zero director colinear to u. Change $t = 2 + 3s$ travels all $\mathbb{R}$.
6. **D.** No value of the parameter is suitable: the line is strictly parallel to the plane.
7. **A.** Each point on the right checks the equation of the plane. The intersection assembly is all the Right.
8. **C.** The same point can be reached for two different values of the parameters. must be checked.
9. **B.** Both constraints leave a free parameter; take $x = t$ imposes $y = 1 - t$ and $z = 2$.
10. **D.** The coefficient of y An equation can still define a plane when a coefficient is zero.

True/false. False: they can be non-coplanar. False: they have normal colinear but Incompatible constants; the planes are strictly parallel. True: the normal straight from M Cut the plan into one point.

Corrected applications and support

Application 1. Set t=2+3s; substitution gives the three coordinates of the second setting and travels $\mathbb{R}$. For (1;3;2), t=1 satisfies the three coordinates.

Application 2. Equation $x + 2y - z = 1$. The proposed item gives 1 $+ 2 - 2 = 1$, therefore belongs to the possible directions $(2; -1; 0)$ and $(1; 0; 1)$, orthogonal to normal and non-collinear.

Implementation 3. d_1 cut in (1.0;1); d_2 is strict parallel; d_3 is included. t=1, 0=1 and 1=1, respectively.

Implementation 4. The equations are t=1+s, t=1-s and 0=s. So s=0, t=1, point (1;1;0). Solve two equations can produce candidates; only the third check concludes.

Application 5. $d_m : (t; mt; m + t)$. The equation becomes $2mt - m = 1$. If m=0, it is impossible: straight strict parallel. Otherwise $t = (m + 1)/(2m)$ provides a single point. To be perpendicular, $(1; m; 1)$ (1.2;-1); the first co-ordinated requires a ratio 1, incompatible with the Third. No m.

Implementation 6. x=y=t and z=3-2t, so straight $(t; t; 3 - 2t)$, point (0;0;3), direction (1;1;-2). meets both equations; the director is orthogonal to both normal (1;1;1) and (1;-1;0).

From system to geometry

To predict the three cases using the figure, then read equations 0=1 and 0=0 as useful results. To differentiate, provide the substitution to the student who mastered the idea but gets lost in the calculation.

Conduct and correction

Benchmarks for the teacher

Use two different parameters to compare two lines. Check the three coordinates, even if two are sufficient to calculate candidates. A Cartesian equation alone generally describes a plan in space; it does not describe a right. Reinvest the projected chapter 11 for give meaning to system resolution.

Corrected activity

1. A point of d_1 a the coordinates (t,t,t). It belongs to the plan if and only if 3 $t=3$, either $t=1$. The intersection is therefore the only point $(1,1,1)$.

2. For a point of d_2, $x+y+z=3+t-t+0=3$ whatever t. All points on the right are in the plan: $d_2 \subset \mathcal{P}$.

3. For a point of d_3, $x+y+z=t-t+0=0$. Equation $0=3$ has no solution: the right is dissociated from the plan. $(1,-1,0)$ is orthogonal to normal $(1,1,1)$; it is parallel to Plan.

4. After substitution, one obtains an equation with one solution, one identity, respectively. and a contradiction. These three cases correspond to an intersection a point, inclusion and absence of intersection.

E1. $\overrightarrow{AB}=(2,1,-1)\neq\vec{0}$. The right has therefore as representation $(x,y,z)=(1,-1,2)+t(2,1,-1)$, $t\in\mathbb{R}$. Values $t=0$ and $t=1$ give again A and B.

E2. For $A(1,2,3)$, condition $\overrightarrow{AM}\cdot(1,0,-2)=0$ is $(x-1)-2(z-3)=0$. After development, the plan has for equation $x-2z+5=0$. Item A Check it out.

E3. After substitution, $t+2(1+t)+(2-t)=8$ or 2 $t+4=8$, so $t=2$. The only intersection is $(2,3,0)$; check 2 $+2\times3+0=8$.

E4. The second equation gives $x=y$. Set $x=y=t$; the first becomes 2 $t+z=1$. All solutions is $(x,y,z)=(t,t,1-2t)=(0,0,1)+t(1,1,-2)$, a right.

E5. The normals are colinear, the second being double the first. $x-y+z=3$ gives 4 $x-2y+2z=6$, incompatible with the second member 5. The plans are therefore strictly parallel, and non-separating.

Corrected from DM12

1. The coefficients x, y, z give the normal non-zero $\vec{n} = (1, 2, 2)$ in the orthonormal coordinate system.

2. Let's look. H on the normal passing through $M : H = (3 + t, 2t, 2t)$. Membership of $\mathcal{P}$ gives $3 + t + 4t + 4t = 9$, so $t = \frac{2}{3}$. Thus $H = (\frac{11}{3}, \frac{4}{3}, \frac{4}{3})$. We're checking. $\frac{11}{3} + \frac{8}{3} + \frac{8}{3} = 9$ and $\overrightarrow{MH} = (\frac{2}{3})\vec{n}$. So H is the projected, and $MH = (\frac{2}{3})\sqrt{1 + 4 + 4} = 2$.

3. The Normal $(1, 2, 2)$ and $(1, -1, 0)$ are not collinear: the plans cut on a straight line. Like $x = y$, posisons $x = y = s$. The first equation gives 3 $s + 2z = 9$, either $z = \frac{9-3s}{2}$. The intersection is the right $(x, y, z) = (s, s, \frac{9-3s}{2})$, $s \in \mathbb{R}$.

4. Item H a $x_H = \frac{11}{3} \neq \frac{4}{3} = y_H$, so it does not belong to $\mathcal{Q}$ or to the right of intersection.

5. Rating is the co-ordinate z. Its nullity requires $9 - 3s = 0$, therefore $s = 3$ and $K = (3, 3, 0)$. Then $MK = 3 > 2 = MH$. This result is consistent with the property of the project: among all points of the plan, H is closest to M.

Corrected work in the manual or an anal

1. True: the right passes through B and its policy vector $(-2; -1; 3)$ value $-\frac{1}{2}\overrightarrow{AB}$.

2. False: $\vec{n} \cdot \overrightarrow{OB} = 15 - 4 - 1 = 10 \neq 0$.

3. False: for $k = -4$ and $s = \frac{5}{2}$, both representations give $(11; 12; -14)$. The rights are cut, so they're coplanar.

4. True: $d(C, \mathcal{P}) = \frac{|2-(-1)+2+1|}{\sqrt{1+1+1}} = 2\sqrt{3}$.